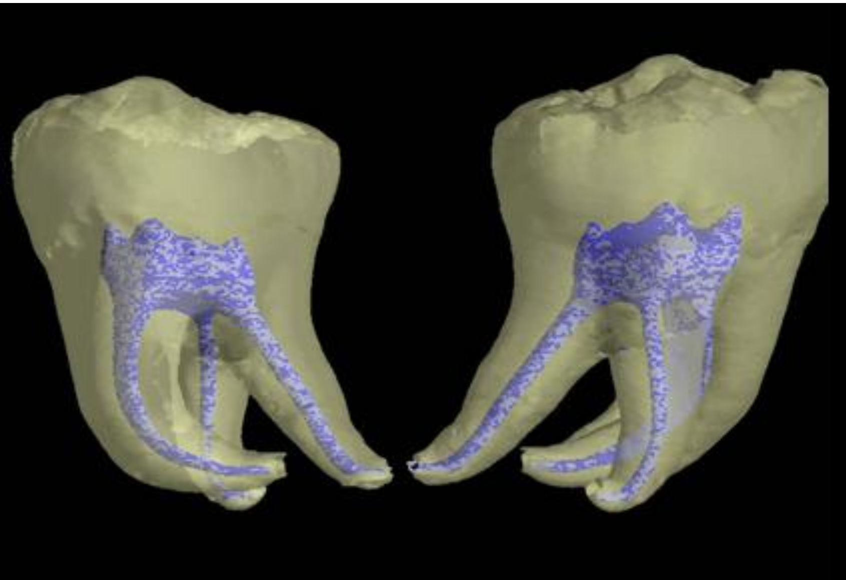
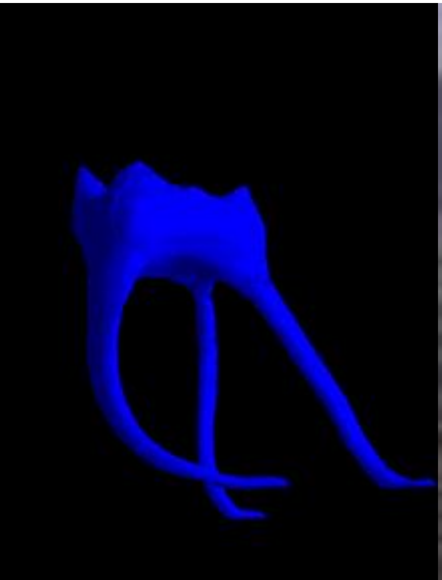
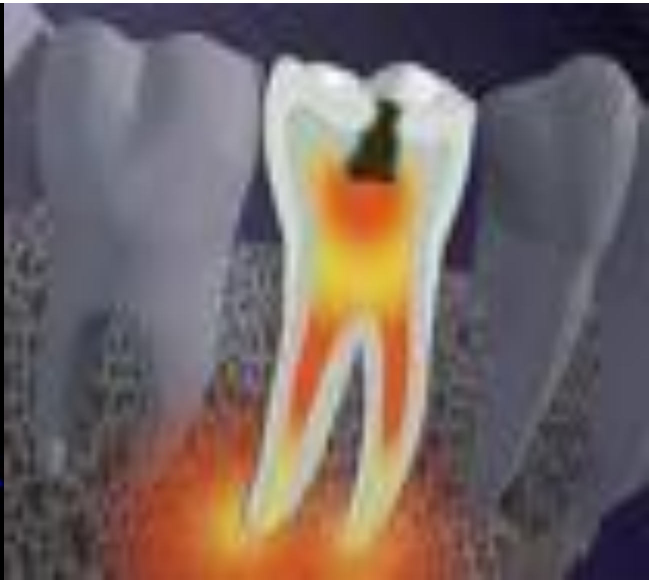
PULP

Part-1 and 2

DR SAJDA GAJDHAR







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By the end of this lesson, students should be able to:

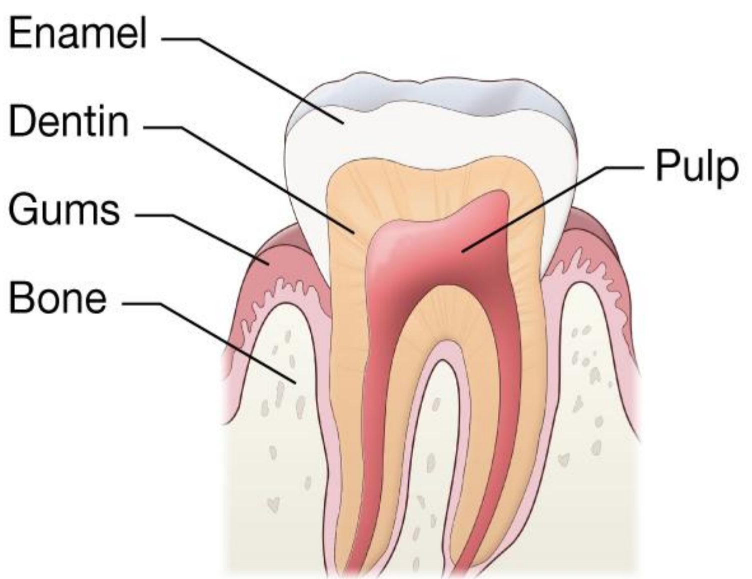
1. Define pulp and its types.

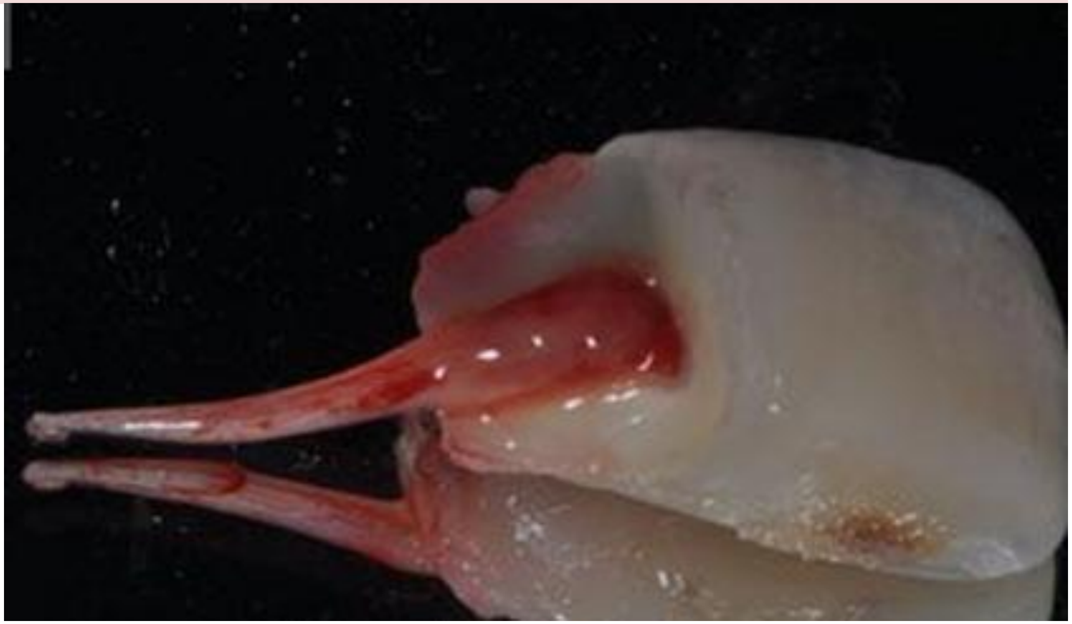
2. Discuss functions and histology of pulp.

3. Describe age changes in pulp.

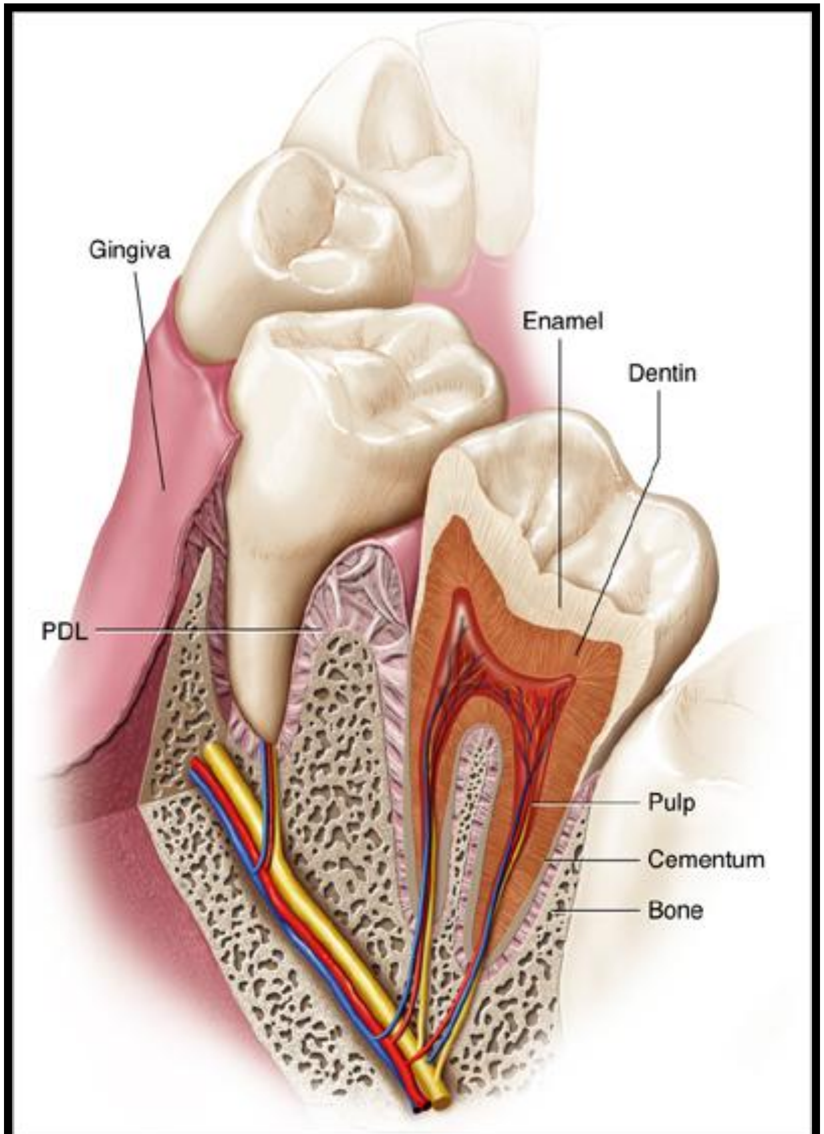
Dental Pulp

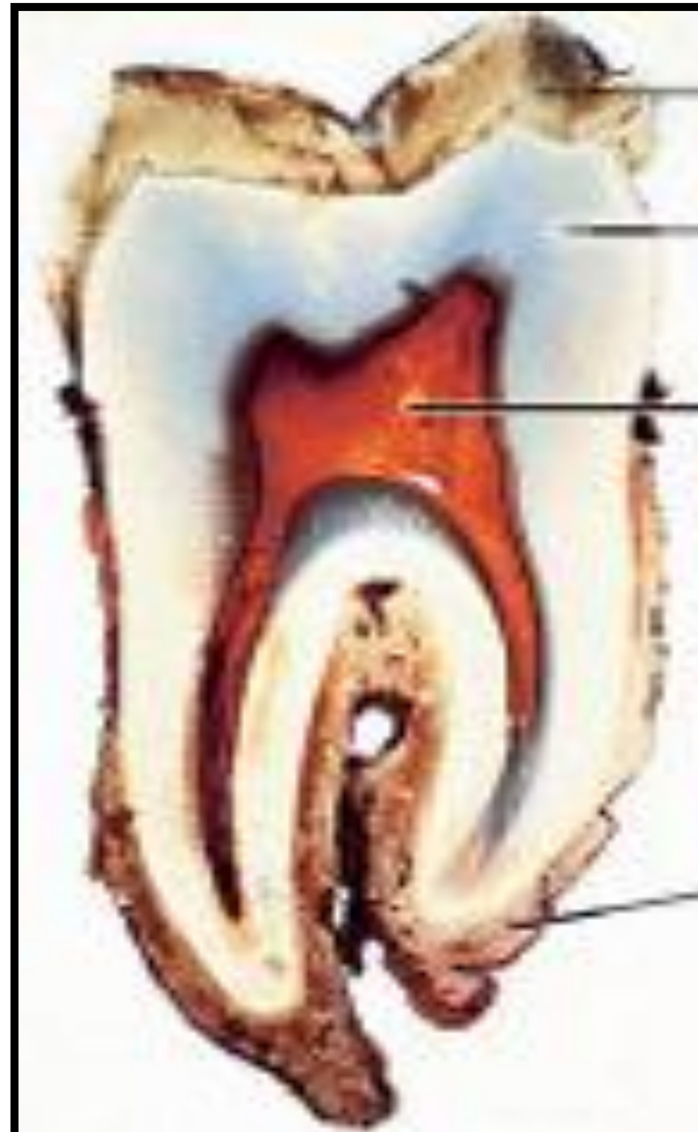
- **Dental pulp is the soft connective tissue located in the central portion of each tooth**.
- **Definition:** Dental pulp can be defined as a richly vascularized and innervated connective tissue of mesodermal origin enclosed by dentin with communications to the periodontal ligament.





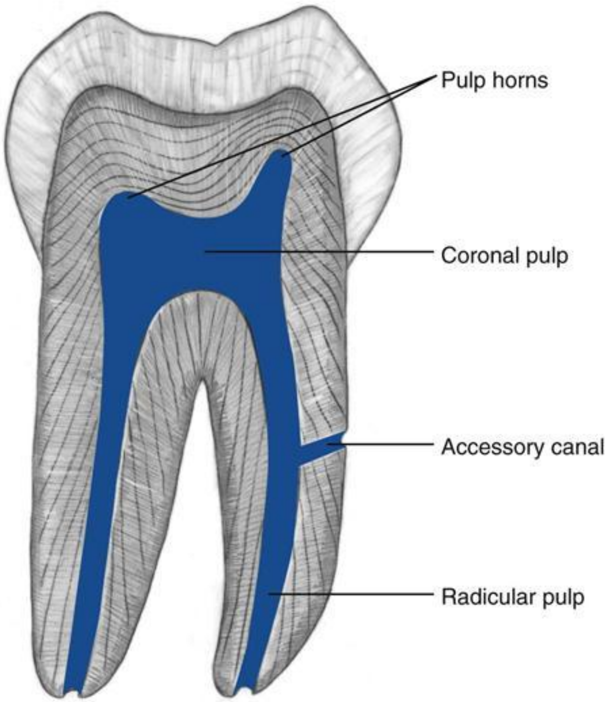
ANATOMY OF PULP



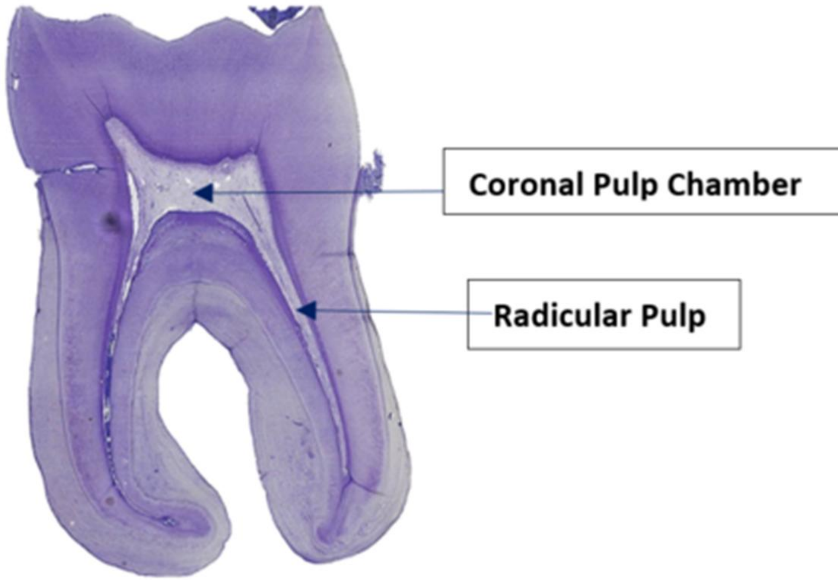


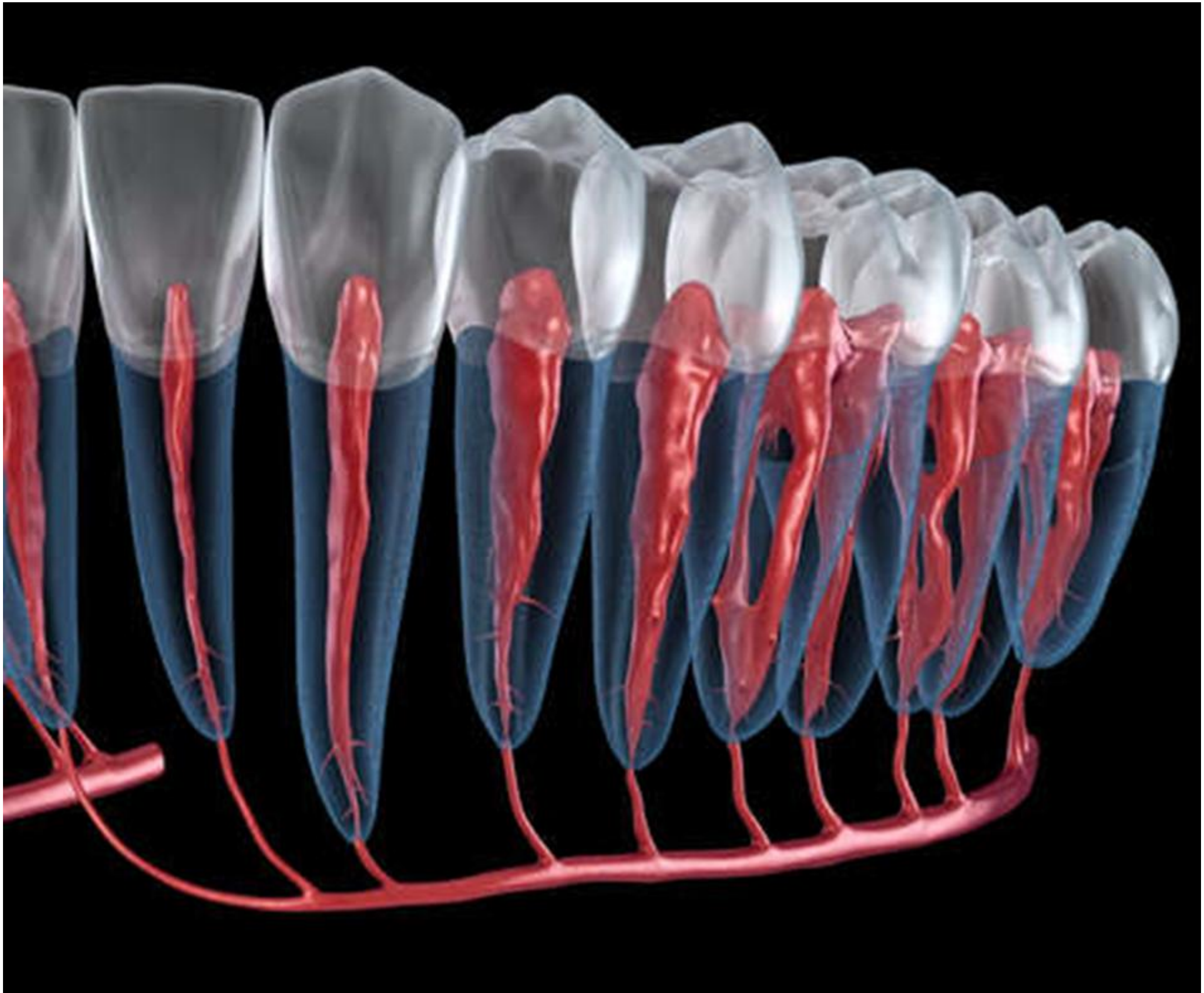
ANATOMY OF PULP

- The pulp which is present in the crown portion of tooth is called as ***Coronal Pulp***
- The pulp which is present in the root portion of tooth is called as ***Radicular Pulp***

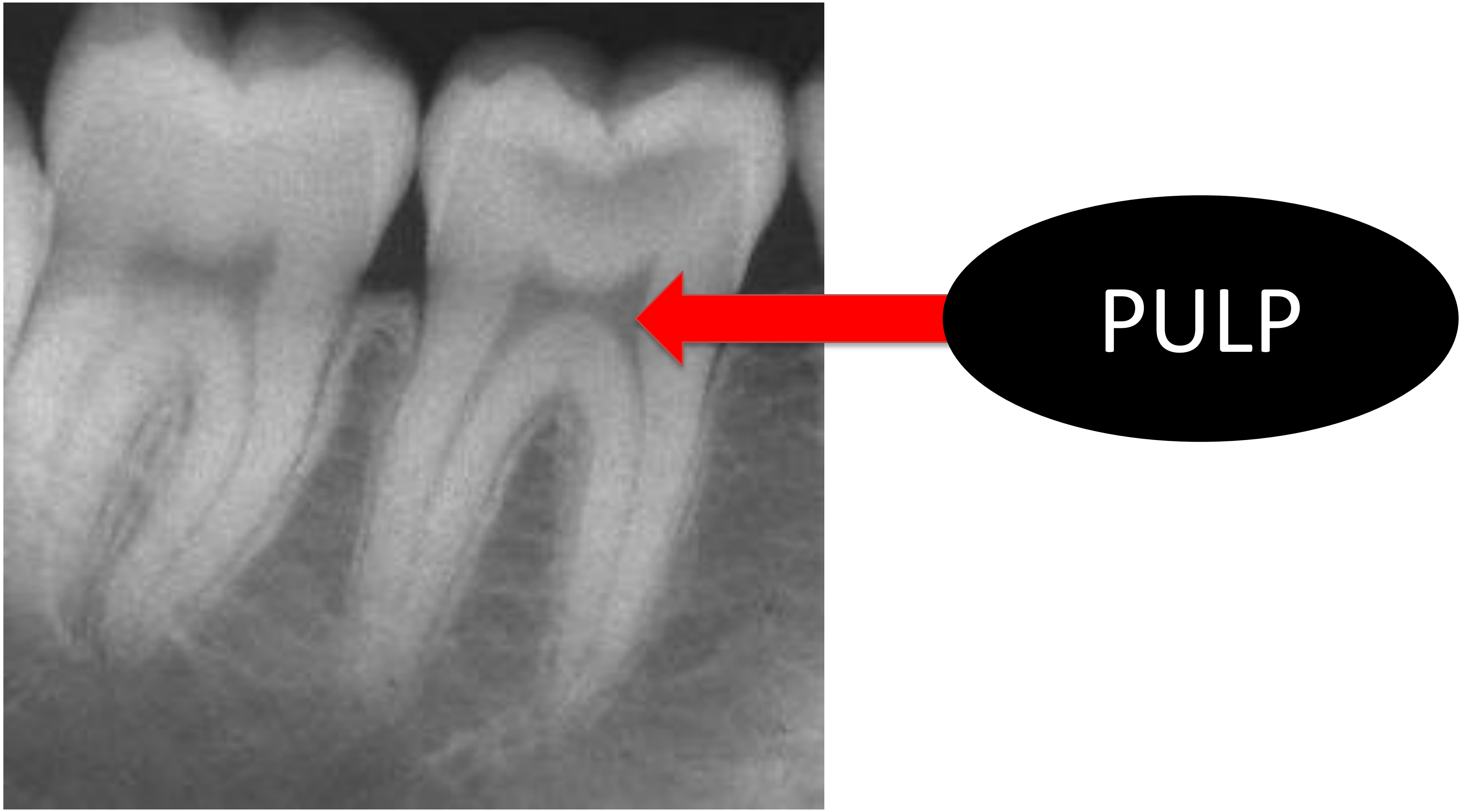


- Pulpal extensions into cusps are known as '***Pulp horns***'
- This portion of tooth is continuous with the periapical connective tissue with ***apical foramen***

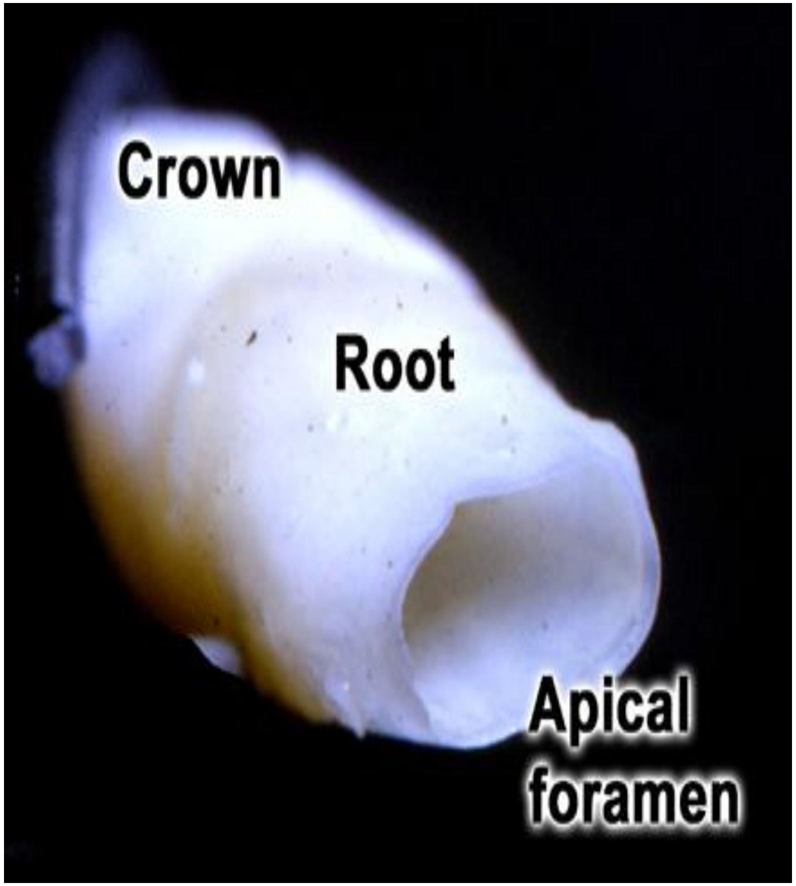




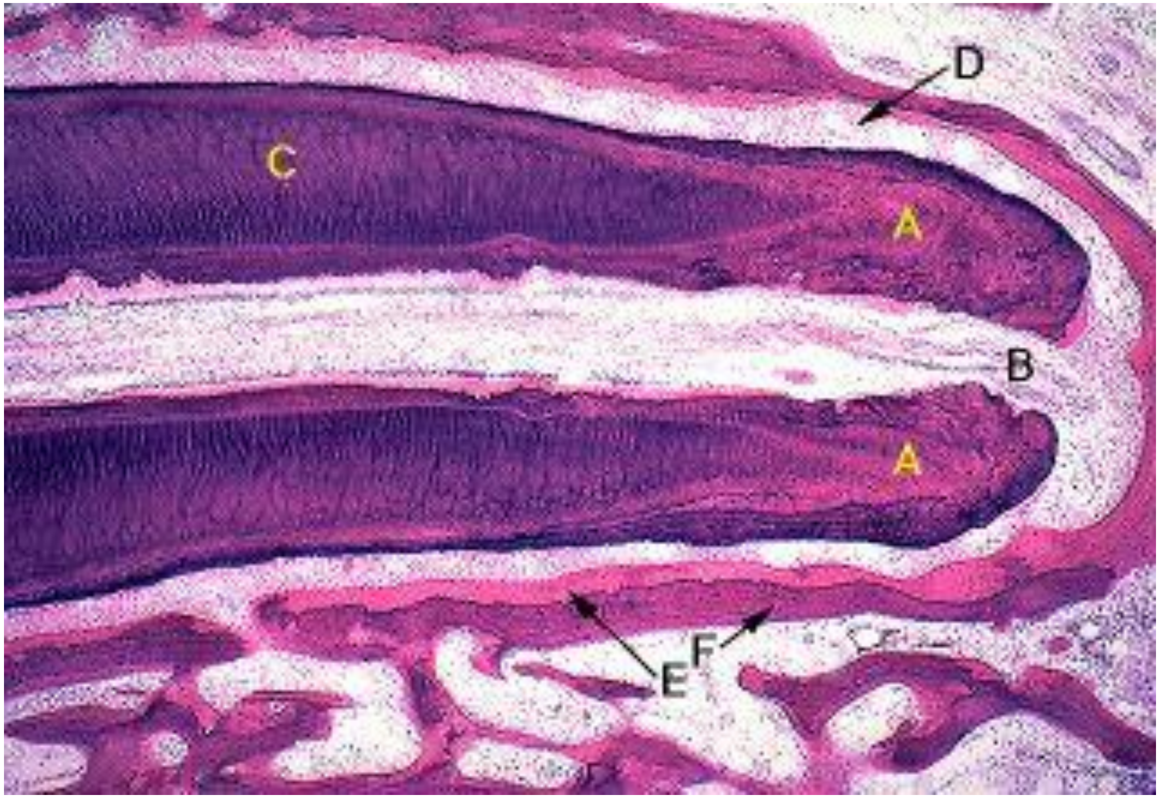
IOPA OF PULP



Apical foramina

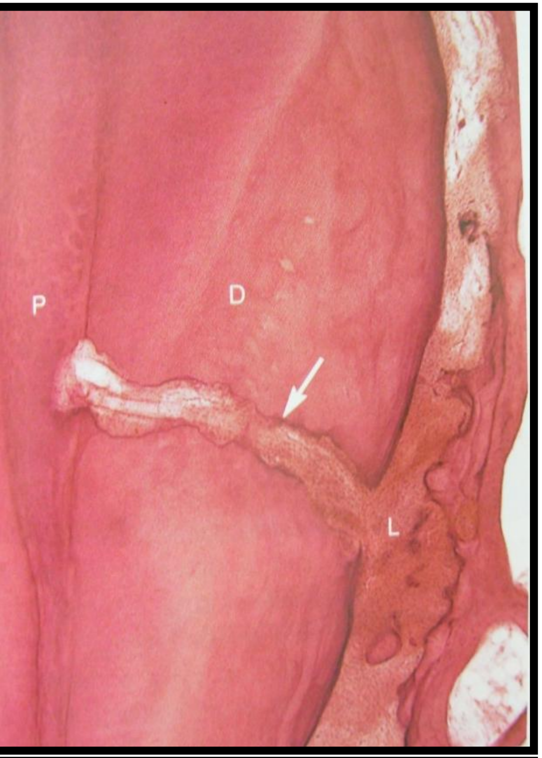


- **Small opening at the root apex through which enter and exist vascular, lymphatic and nerve elements.**



A - cementum
B - apical foramen
C - dentin

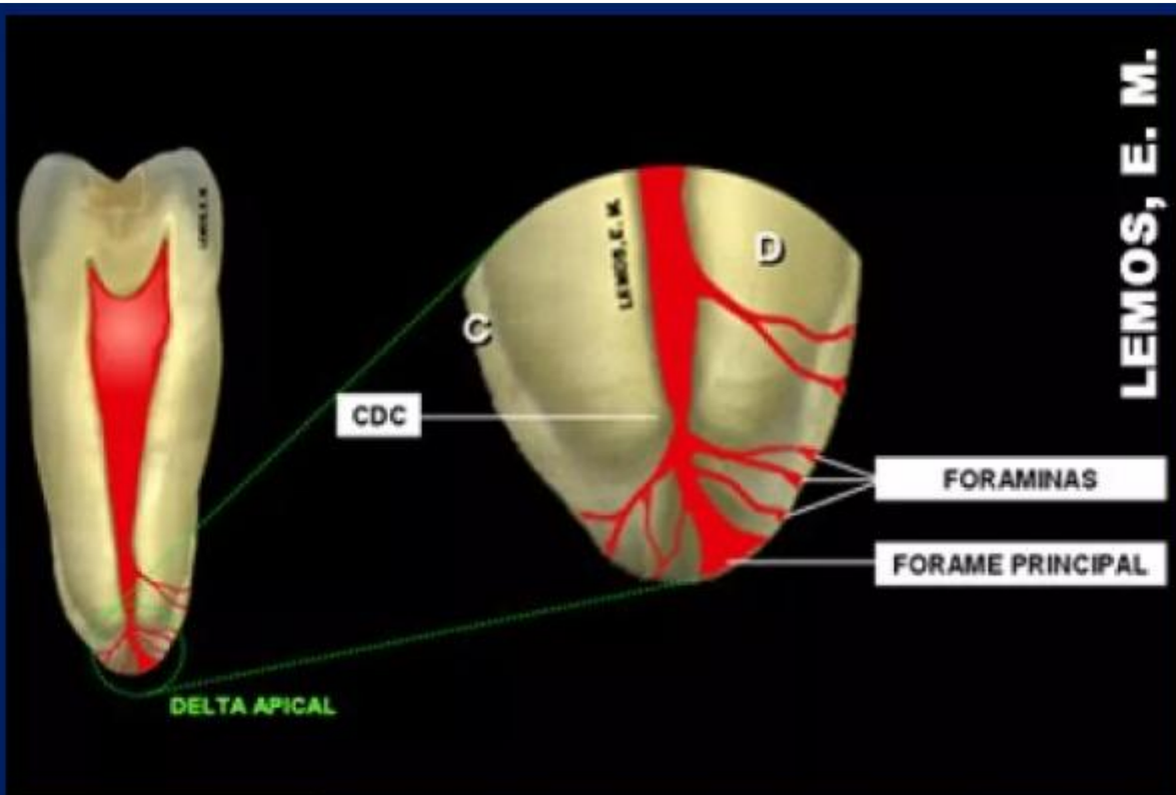
D - periodontal ligament
E - osteoid
F - alveolar bone

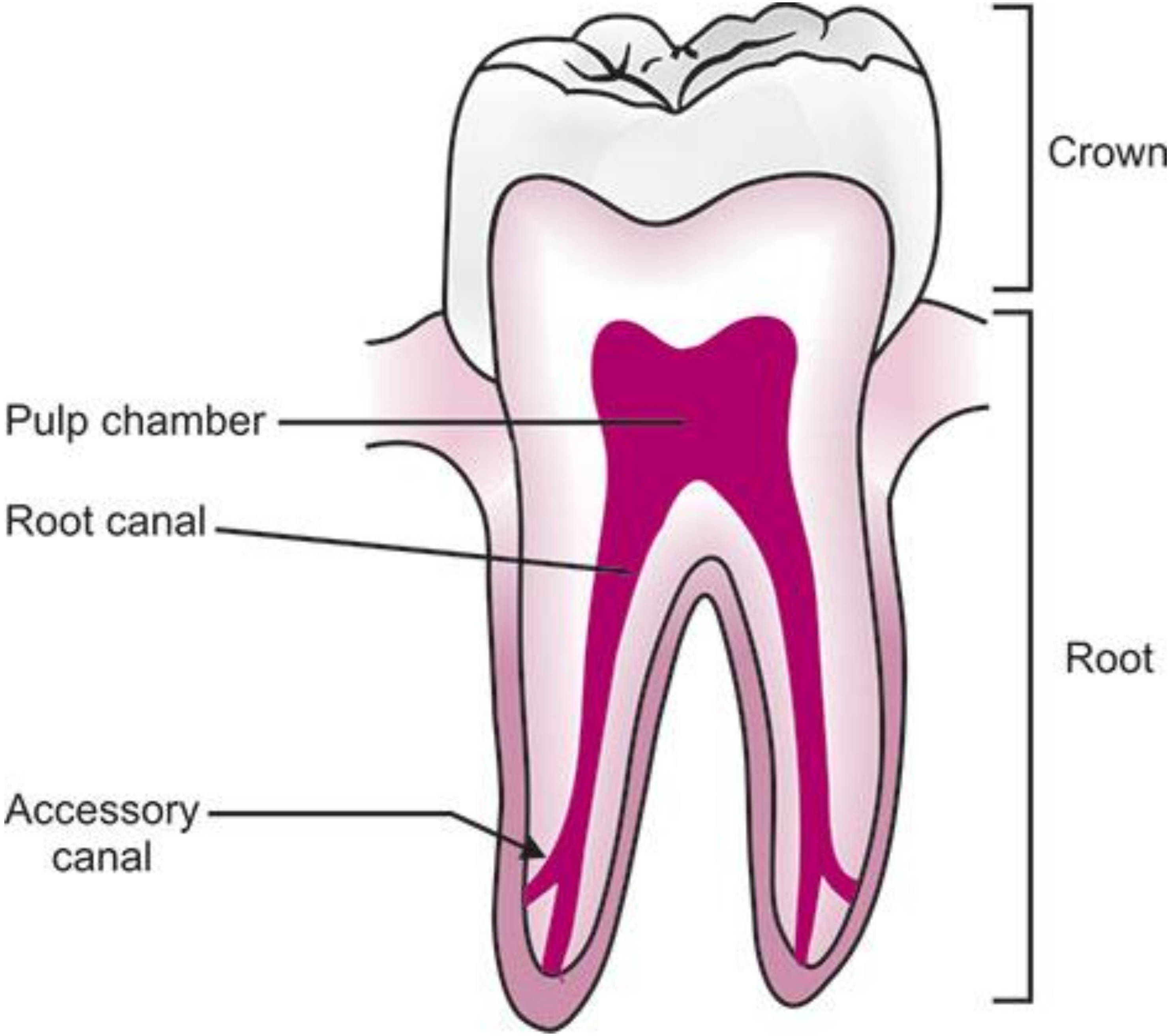


Accessory canals

- **Accessory canals** are small canals leading from the radicular pulp laterally through the root dentin and cementum to the periodontal ligament.
- Most numerous **in the apical third of root.**

- **Clinically significant in spread of infection.**


LEMOS, E. M.
CDC
C
D
FORAMINAS
FORAME PRINCIPAL
DELTA APICAL


Crown
Pulp chamber
Root canal
Root
Accessory
canal

Functions Of Dental Pulp

1. **Formation of Dentin**:
    1. The primary role of odontoblasts in the dental pulp is the formation of dentin. These cells secrete dentin matrix during tooth development and continue to form secondary dentin throughout life.
    2. The pulp is also responsible for the formation of **tertiary dentin** (also called reparative dentin), which forms in response to stimuli such as caries, trauma, or dental procedures.
2. **Nutrient and Oxygen Supply**:
    1. The pulp's blood vessels deliver essential nutrients and oxygen to the tooth, particularly to the dentin, which is not directly supplied by blood vessels. The pulp is thus crucial for maintaining tooth vitality.

3. Sensory Function:

1. The dental pulp is highly innervated by sensory nerve fibers, primarily from the **trigeminal nerve**. These fibers allow the tooth to detect various stimuli, such as temperature (hot and cold), pressure, and pain. Pain sensation from the pulp is intense due to the small volume of the pulp chamber and the density of nerve fibers.
2. **Pain as a Protective Mechanism**: The pulp's sensory nerves act as a warning system for potential damage, such as infection or trauma.

4. Defensive and Reparative Function:

1. The pulp plays an important role in defending against infection and injury. When the tooth is subjected to caries or trauma, the pulp activates its **immune response**, which includes an increase in the number of immune cells (like macrophages and neutrophils).
2. **Reparative Dentin Formation**: In response to injury, the pulp can stimulate the formation of tertiary dentin by odontoblast-like cells to shield the pulp from further damage.

5. Regenerative Potential:

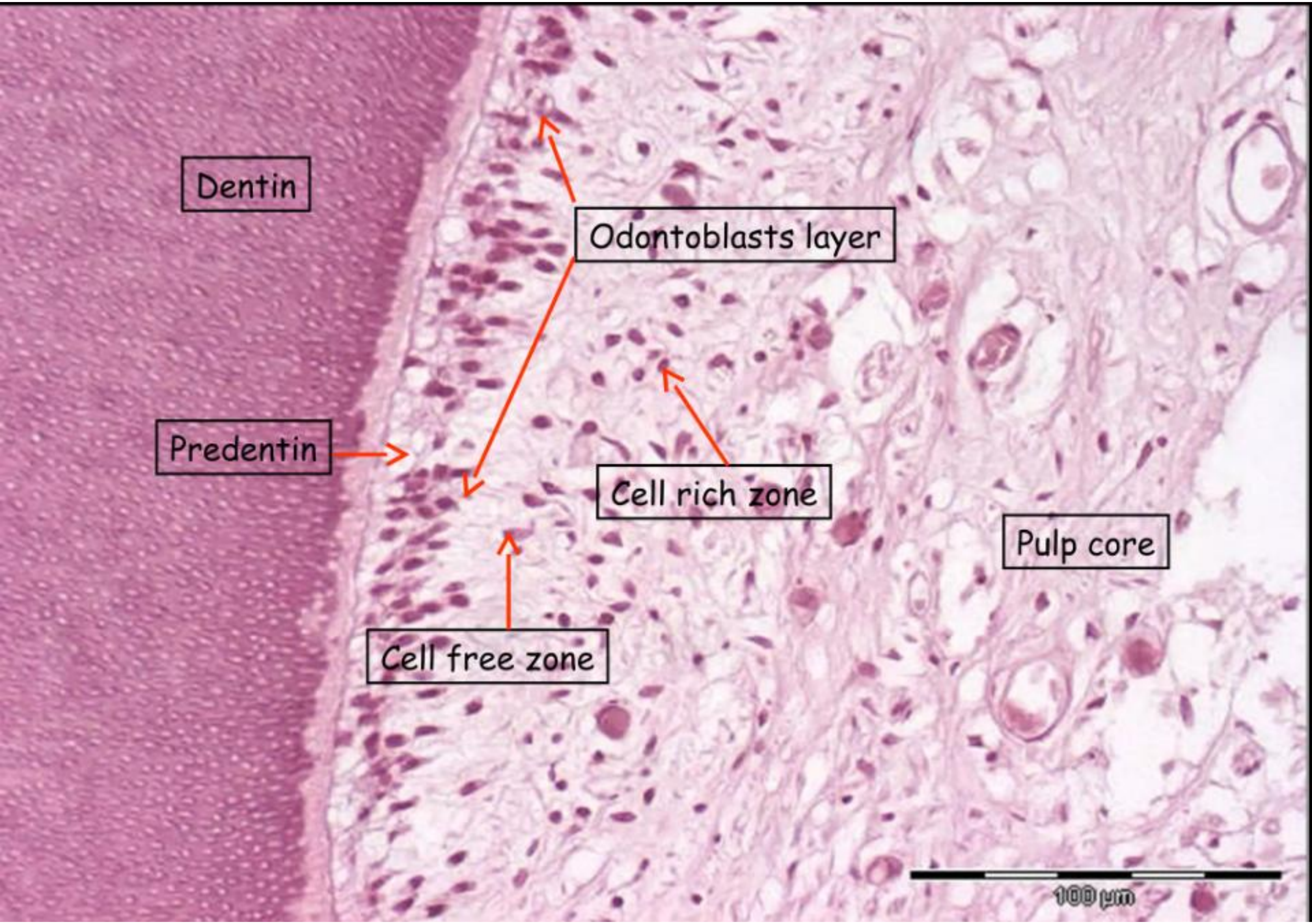
1. The dental pulp contains **mesenchymal stem cells** that have the potential to differentiate into odontoblast-like cells in response to injury or damage. This regenerative ability is of growing interest in dental research, particularly for potential regenerative therapies.

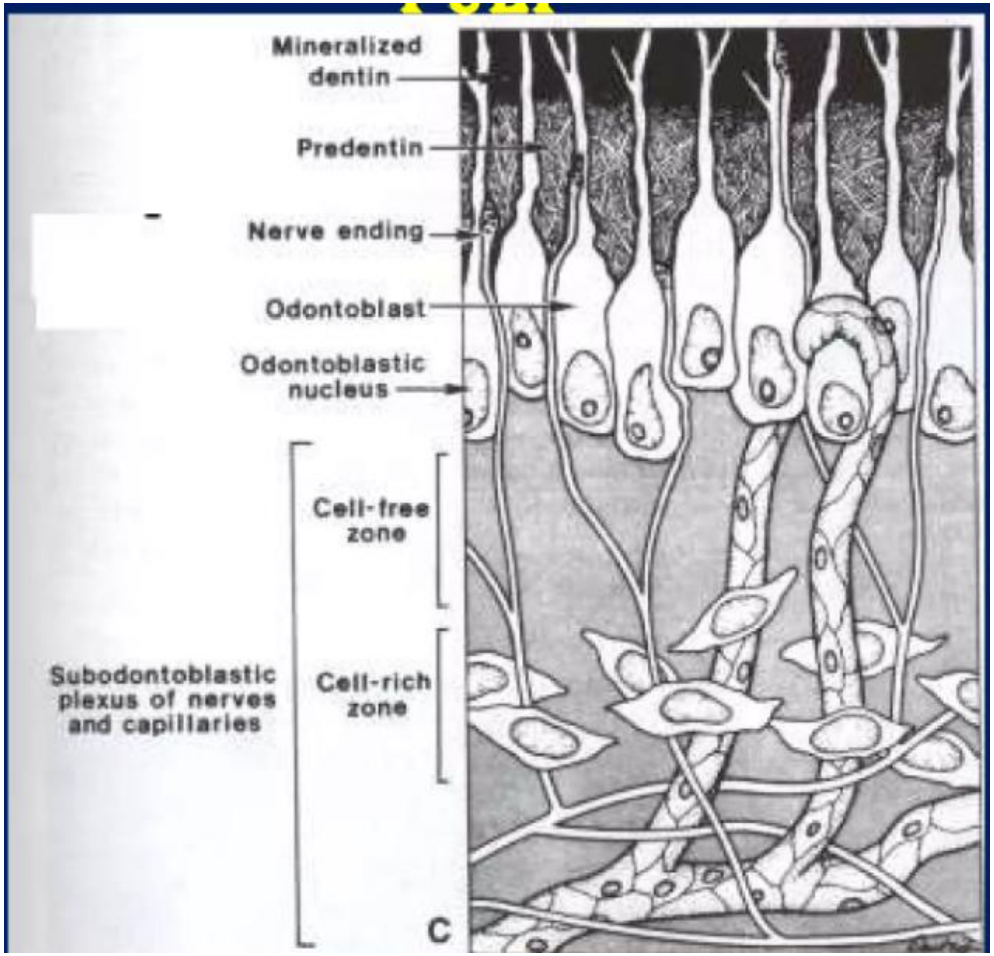
HISTOLOGICAL ZONES

- **Pulp shows two regions –**
- **Pulp at periphery** – composed of 3 zones-

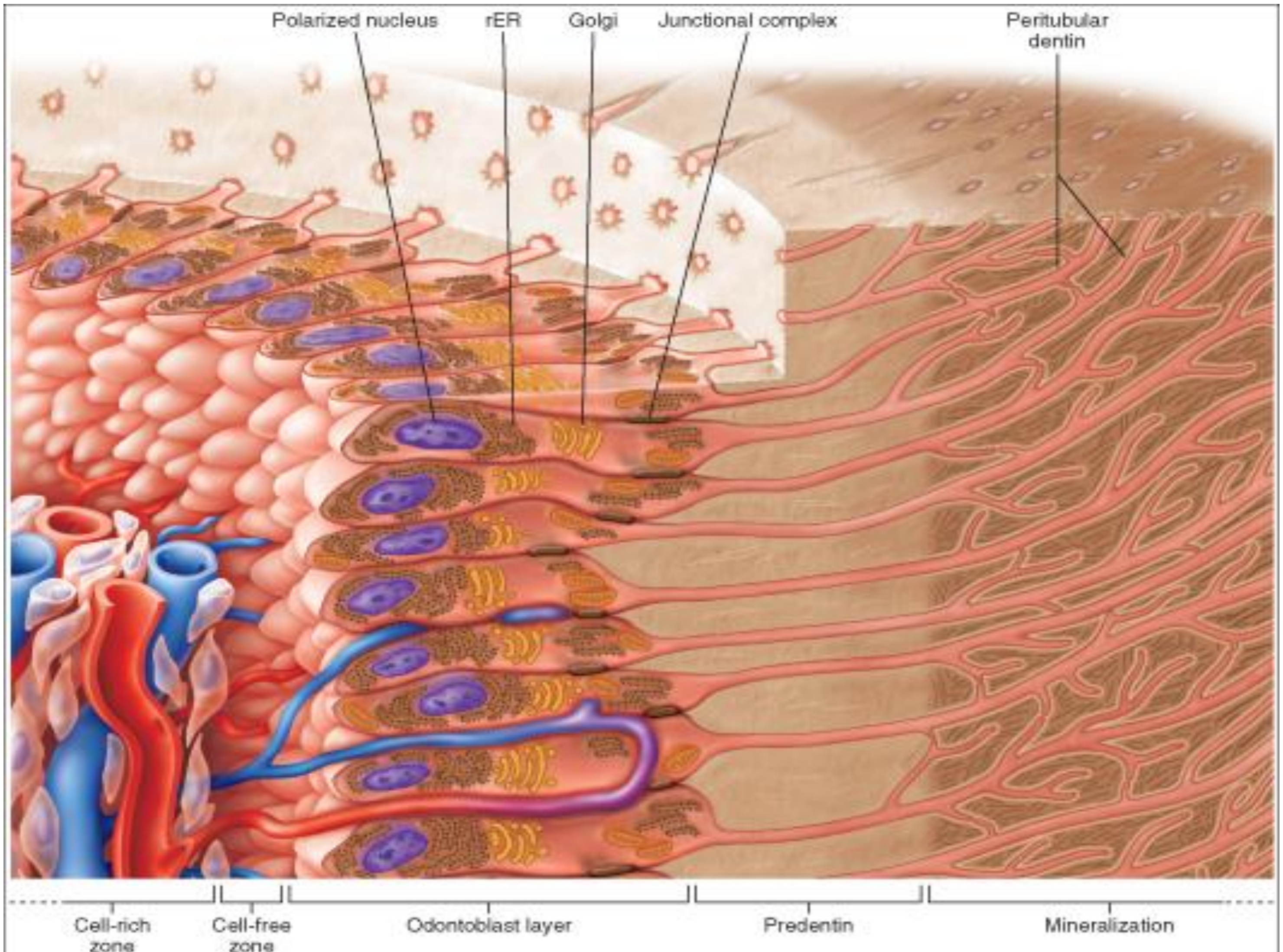
1. **Odontoblastic Zone** – Outermost zone adjacent to pulp dentin border.
2. **Cell- free zone** – middle **cell free zone or weils zone** is the space that is seen between odontoblastic zone and cell rich zone.
3. **Cell- rich zone** – It is restricted to coronal region.

- **Pulp at center** – Central region of both coronal and radicular pulp contains **large nerve trunks and blood vessels.**


Dentin
Odontoblasts layer
Predentin
Cell rich zone
Pulp core
Cell free zone
100 µm

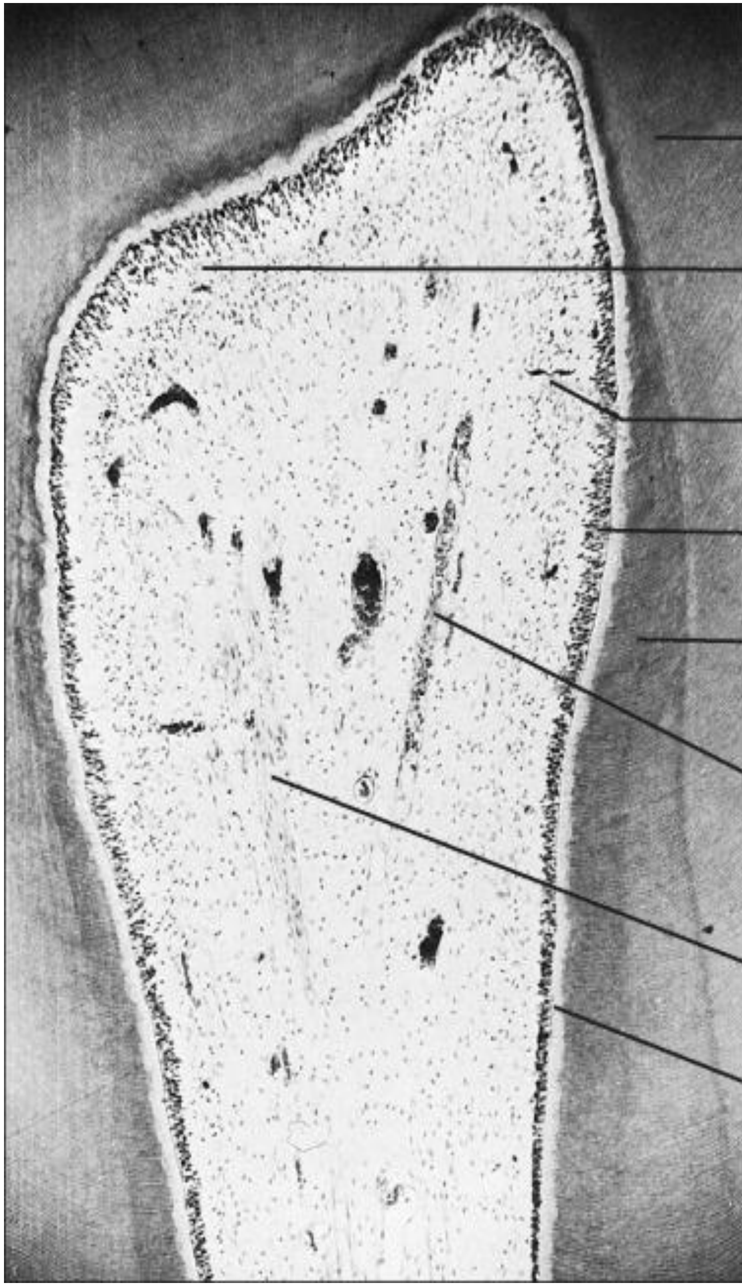

Mineralized dentin
Predentin
Nerve ending
Odontoblast
Odontoblastic nucleus
Cell-free zone
Cell-rich zone
Subodontoblastic plexus of nerves and capillaries
C

Odontoblastic Zone



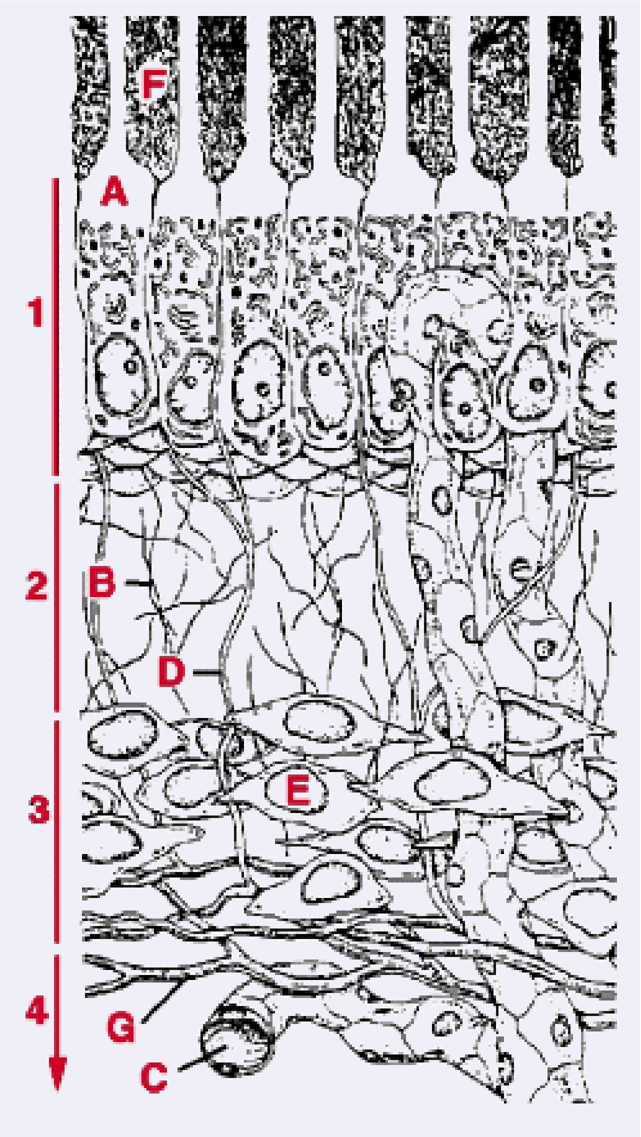
- Odontoblasts, dentin forming cells arranged in palisading manner ,the second most prominent cell in the pulp, reside adjacent to the predentin with cell bodies in the pulp and cell processes in the dentinal tubules.
- Approx. 5-7 µm –diameter

  25-40 µm- length
- Though these cells are arranged in single row sometimes they crowd and appear **pseudostaratified.**



Various morphologic variations occurs

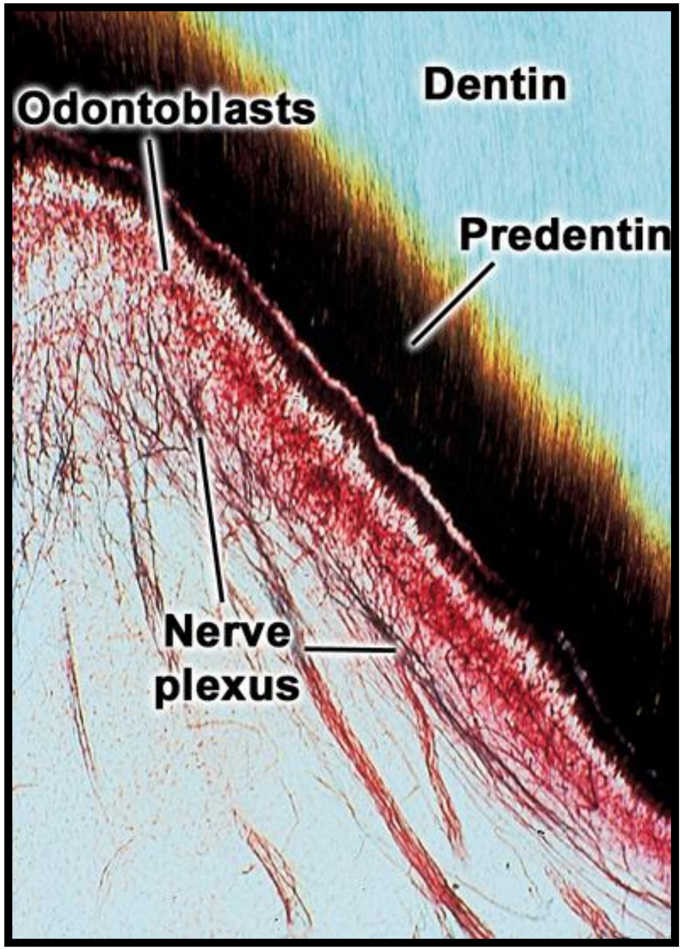
- Crown portion – tall columnar
- Middle of root - low columnar
- Root portion - shorter or more/less cuboidal
- Apex - flattened



F
A
1
2
B
D
3
E
4
G
C

Cell- free zone of Weil

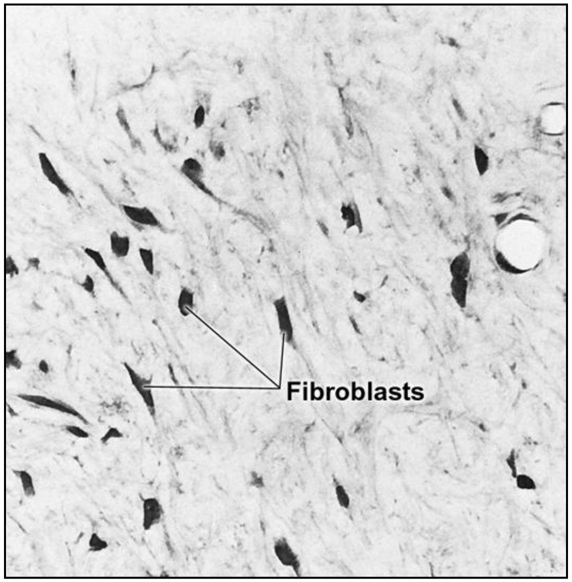
- Immediately next to the Odontoblastic layer in the coronal pulp.
- Often a narrow zone of about 40 μm in width that is **relatively free of cells.**
- **At cell free zone just below the cell bodies of odontoblast in the crown portion plexus of nerve form which is called as NERVE PLEXUS OF RASHCHOW**

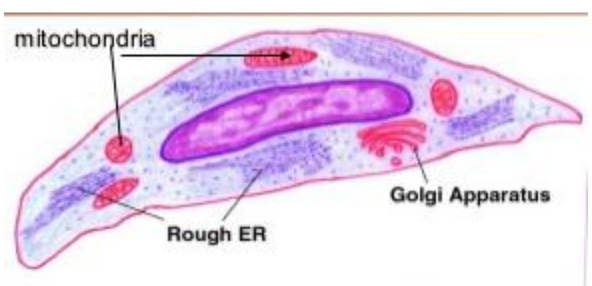


Cell Rich Zone

- The area containing a relatively **high proportion of cells compared with the more central region of the pulp**
- More prominent in the coronal pulp
- **Rich in fibroblast**, Also includes a variable no of <u>macrophages, dendritic cells & lymphocytes</u>

Fibroblast





- Fibroblasts **are the most numerous cell type** in the pulp.
- In **young pulp** they are **Stellate shaped** and extensive process.
- In older pulp they **appear rounded or spindle shaped with short processes and few intracellular organelles**. They are called as **Fibrocytes.**
- Produces collagen fiber (type-I Collagen) and extracellular matrix.
- Function in formation and resorption of the same matrix.

Undifferentiated mesenchymal Cells

- They are found along the pulp vessels, in cell rich zone and scattered throughout the central pulp

- They are believed to **be a totipotent cell** and when need arises they may become odontoblasts, fibroblasts, or macrophages.

- They decrease in number in old age.

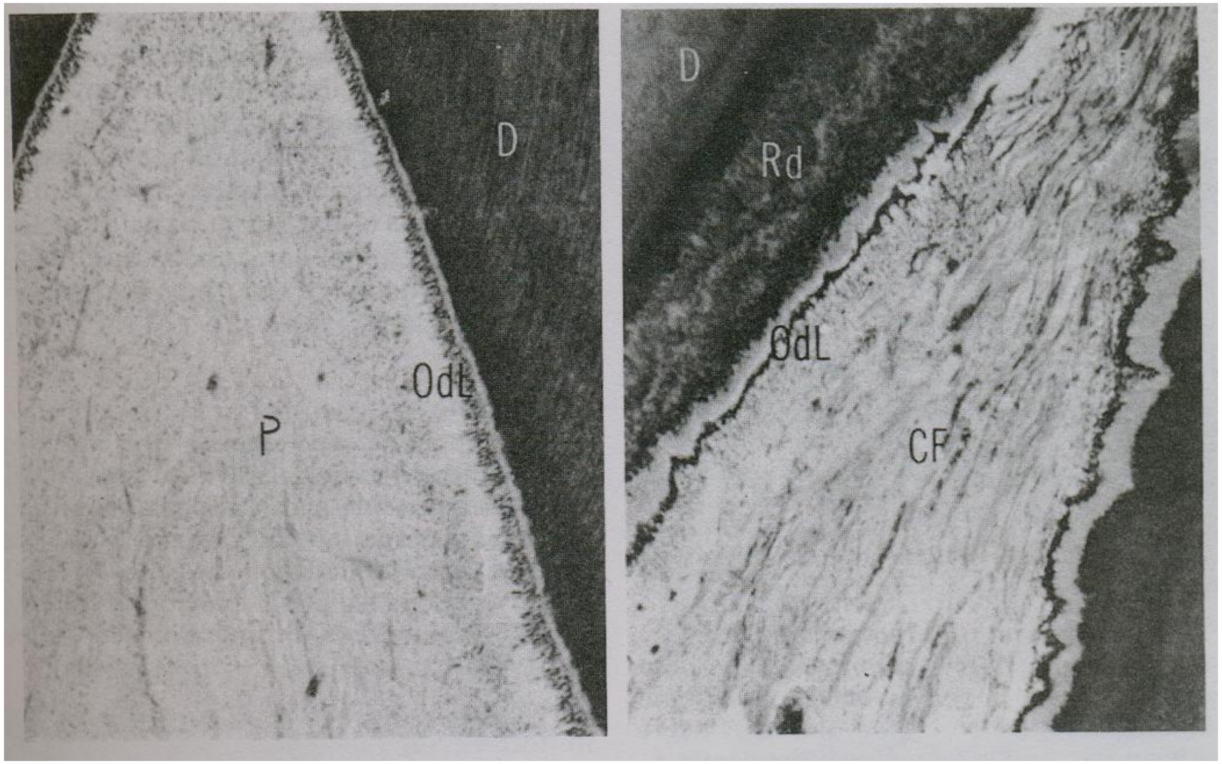
Defense Cells

- Histiocytes or macrphages
- Mast cell , Plasma cell
- Neutrophils, Eosinophilis, Basophils, Lymphocytes,& Monocytes
- Dendritic Cells

Regressive/Age Changes

- Cellular changes
- Fibrosis
- Vascular changes
- Pulp stones
- Diffuse calcifications

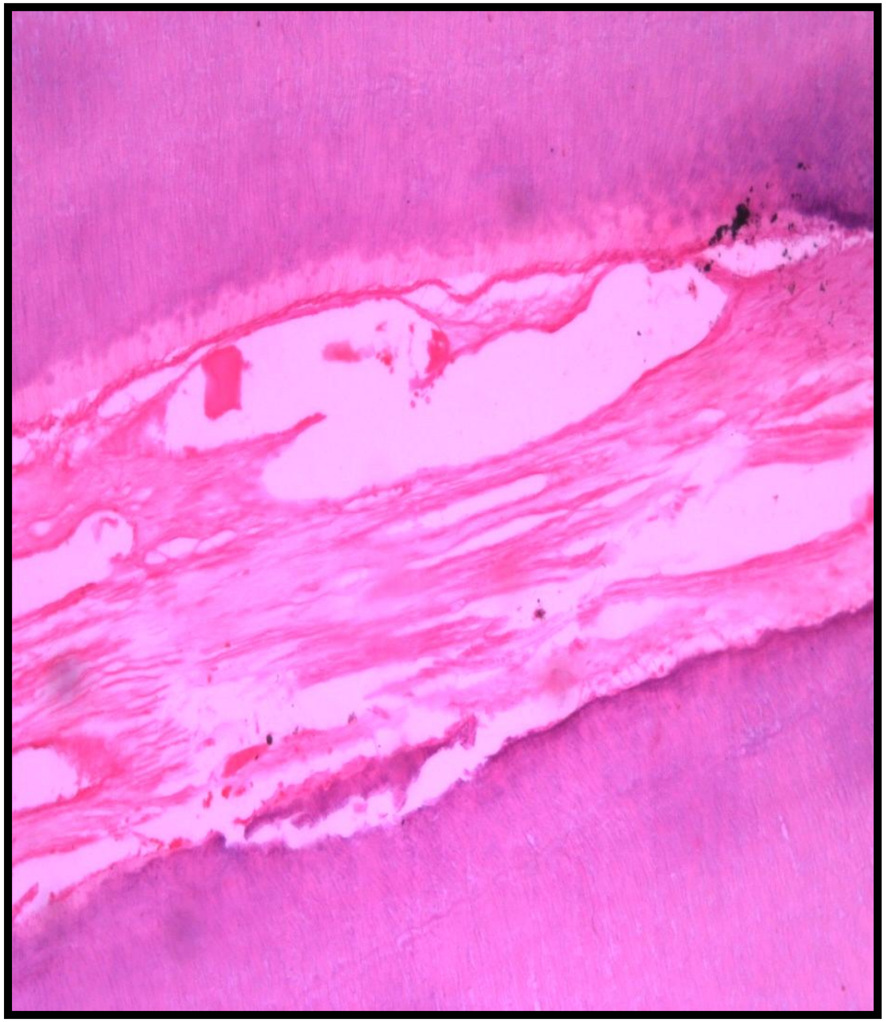
AGING OF DENTAL PULP



- **Young pulp-highly cellular, few collagen fibers**

Aged pulp-abundantly present collagen fibers, deposits of reparative dentin

Fibrosis



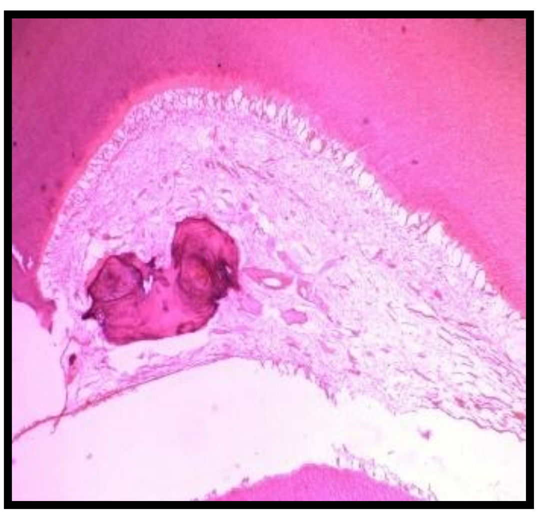
Increased in matured collagen fibers is called fibrosis.

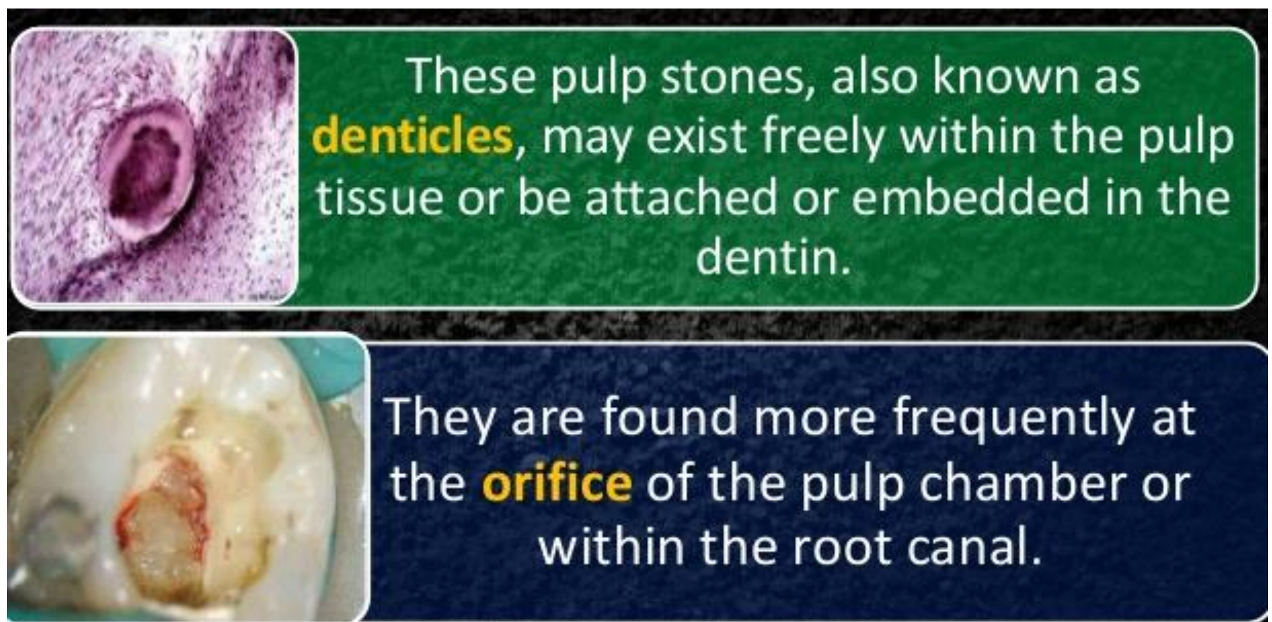
Accumulation of both diffuse fibrillar components as well as bundles of collagen fibers

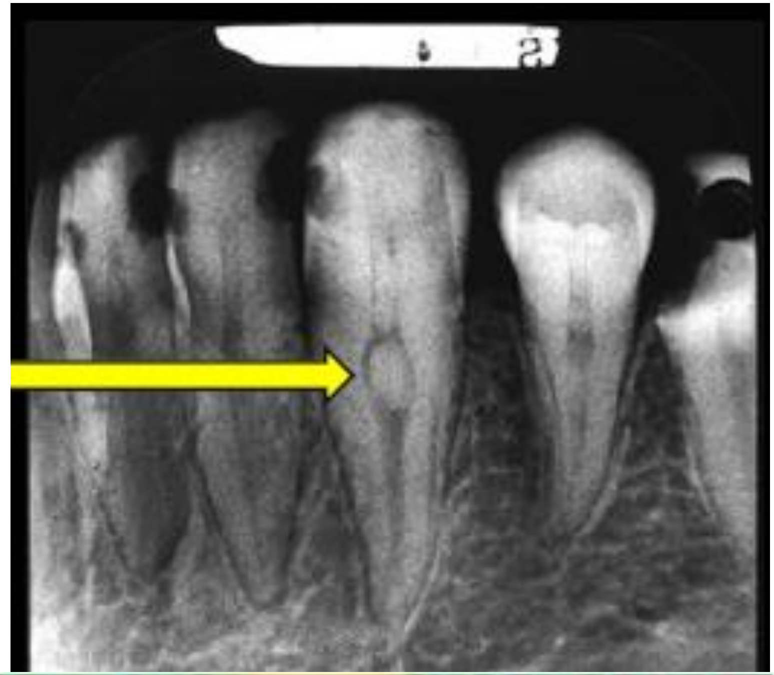
Pulp Stones

Pulp stones/ Denticles:-

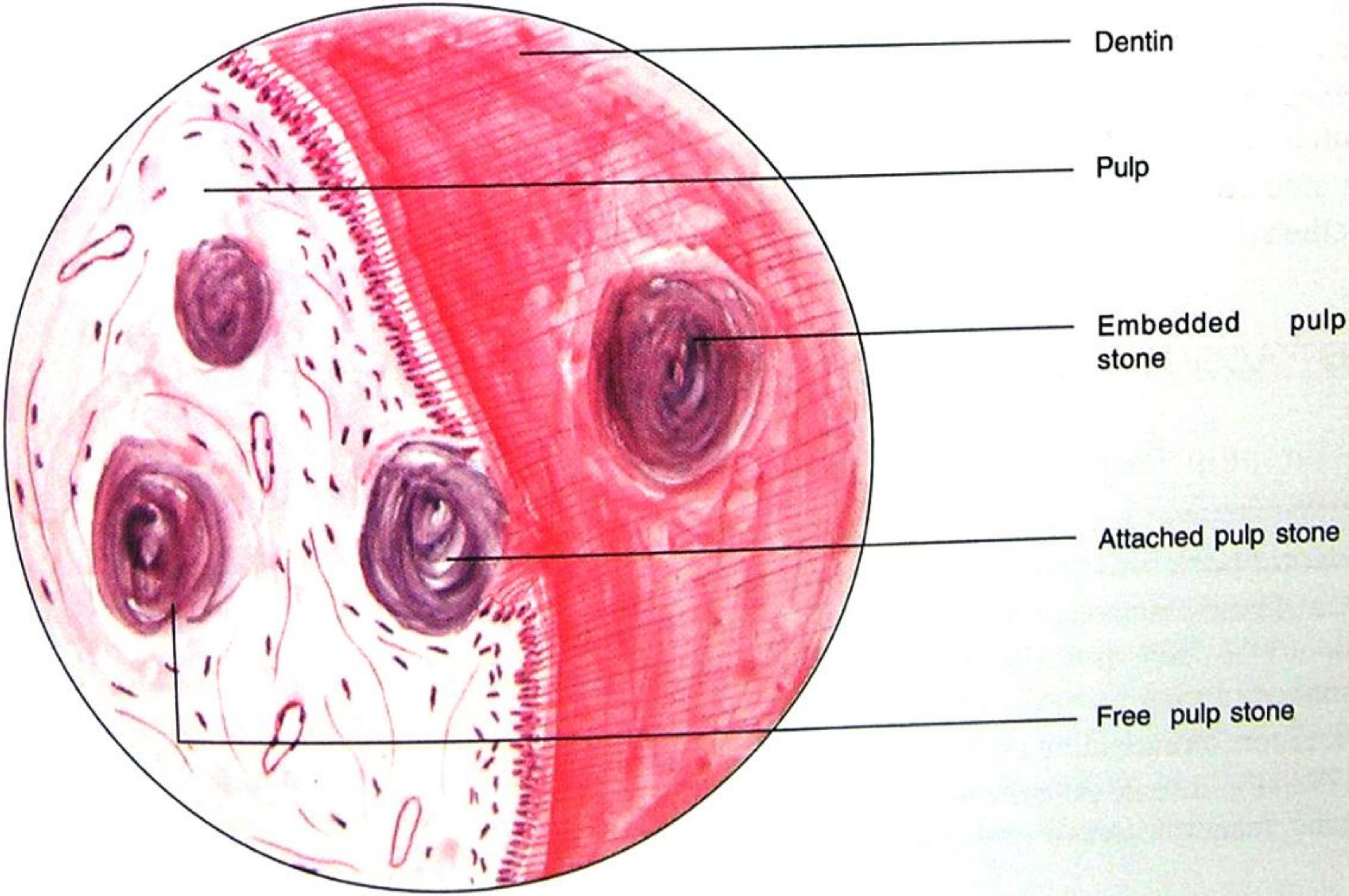
- The nodular calcified masses in coronal as well as radicular pulp are called pulp stones or denticles.
- Asymptomatic
- Functional /embedded unerupted teeth







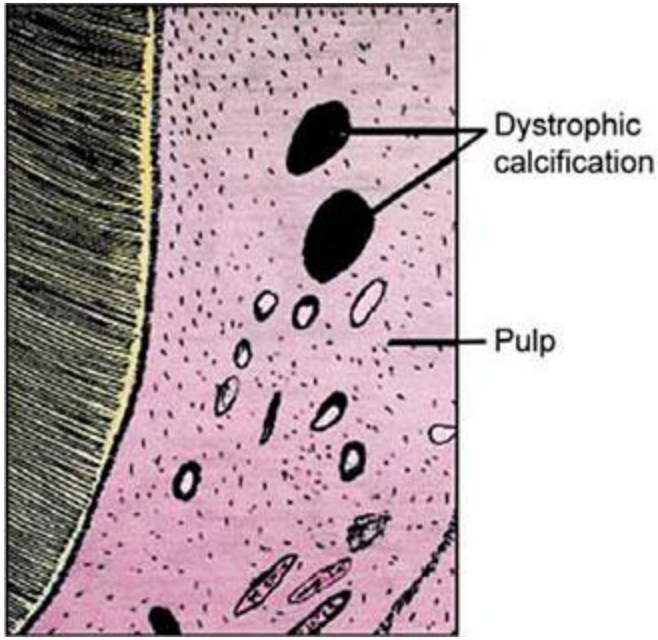
Pulp Stones

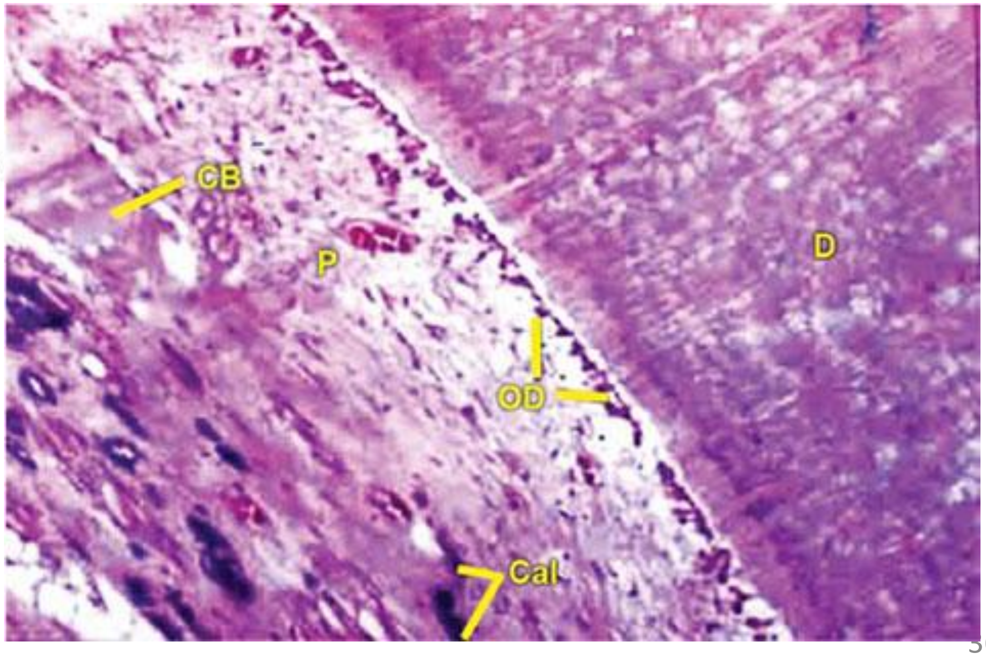

Dentin
Pulp
Embedded pulp stone
Attached pulp stone
Free pulp stone

Pulp Stones

Diffuse Calcification

Diffuse calcifications appear as **irregular calcific deposits** in the pulp tissue, usually following collagenous fiber bundles or blood vessels. It is also called as calcific degeneration.





THANK YOU

"Excellence
is not being
the best;

it is doing
your best."